

Mohamed Rostom

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PROFESSIONAL SUMMARY

Software Engineer with 4+ years of experience in embedded systems and automation, specializing in high-performance solutions for next-generation embedded and SoC platforms. ISTQB-certified, with strong expertise in C++, Python, Java, and Linux. Skilled in test automation, CI/CD, and Agile delivery, with a proven record of driving efficiency and delivering complex projects in global, cross-functional environments.

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, Java, CAPL, Shell/Bash
- **Operating Systems:** Linux (Embedded & Real-Time), Windows
- **DevOps & CI/CD:** GitLab CI/CD, Jenkins, Git
- **Tools & IDEs:** VS Code, IntelliJ, Vector CANoe/Test Studio, UML, Postman
- **Databases:** SQLite, PostgreSQL, MySQL
- **Collaboration & ALM Tools:** Jira, Confluence, GitHub, GitLab, Core ALM, IBM DOORS, STARC
- **Methodologies & Standards:** Agile (Scrum, TDD), ASPICE, V&V

PROFESSIONAL EXPERIENCE

Software Automation Test Engineer

Mercedes-Benz ADAS Project (via Luxoft GmbH)

Jul 2024 – Present

- **Contributing in automation and CI/CD initiatives** for the Mercedes-Benz **ADAS** project by designing and implementing pipelines in **GitLab CI/CD**, streamlining build, integration, and validation of automotive software.
- Developed reusable **Python libraries** and GitLab pipelines to automatically package, version, and publish components, enabling consistent reuse across ADAS ECU projects and reducing delivery effort.
- Designed and implemented **test automation frameworks** using **Python, CAPL, and Vector CANoe** for ECU validation of ADAS functions, including sensor fusion, environment perception, and driver assistance features.
- Maintained and extended **HIL (Hardware-in-the-Loop)** environments for ADAS ECU testing, ensuring robust validation of real-time sensor inputs and vehicle communication protocols.
- Selected as a **Resident Engineer on-site at Mercedes-Benz**, collaborating directly with OEM engineers to debug integration issues, optimize automation workflows, and ensure compliance with **ASPICE** standards.
- Authored and maintained detailed **technical documentation** (Jira/Confluence) covering CI/CD workflows, test frameworks, and release pipelines, ensuring traceability of ADAS requirements through V&V processes.
- Collaborated with cross-functional R&D teams to harmonize validation strategies, and automation frameworks in ADAS projects.
- Key technologies: **Python, CAPL, GitLab CI/CD, Jira, Confluence**

Software Engineer

Valeo

Nov 2022 – Jul 2024

Client

Nissan, Volkswagen

- Extensive experience in **embedded software design, development, implementation, and testing** on Linux platforms and real-time systems.
- Strong knowledge of **embedded Linux programming, system-level software architecture, and best practices** across the full software development lifecycle.
- Proficient in **debugging complex software and hardware interactions**, ensuring robust and reliable solutions.
- Skilled in **product testing, validation, and quality assurance** for automotive and embedded applications.
- Hands-on expertise in developing and maintaining **software update and configuration management components** for telematics devices using **C/C++ and SQLite**.
- Effective communicator and collaborator with proven ability to:
 - Contribute to projects involving **vehicle-to-vehicle communication, vehicle diagnostics, and remote firmware updates**
 - **Engage with system teams and customers** via email, and online meetings to clarify requirements and align on deliverables
 - Conduct regular **technical and design review meetings** with stakeholders to ensure alignment on requirements, design, and testing
 - Collaborate seamlessly with **software engineers, testers, and project managers** in global teams
- Key technologies used: **C/C++, Python, CAPL, SQLite, Linux**

Validation Engineer

Valeo
Client

Oct 2021 – Oct 2022
Nissan, Renault

- Designed and executed **comprehensive validation test plans** for automotive embedded systems.
- Developed and performed **test scenarios for new products** as well as continuous improvement initiatives.
- Conducted a wide range of testing activities, including **smoke, regression, exploratory, daily verification, negative, usability, functionality, embedded system, and feature interaction testing**.
- Installed and configured **daily software builds** in test environments to enable accurate and timely testing.
- **Analyzed test data, documented results**, and provided clear, actionable reports for stakeholders.
- Executed **Verification & Validation (V&V) procedures** to ensure all requirements were met, tracked results, and communicated status updates.
- Managed **CI/CD daily validation testing** and contributed to the **automation of feature test cases**.
- Maintained **HIL (Hardware-in-the-Loop) testing environments** to ensure robust validation of software components.
- Identified and resolved **blocked test cases**, enabling their inclusion in automated pipelines.
- Created and maintained **detailed test documentation** including test plans, cases, and reports using tools like Core ALM and IBM Rational DOORS.
- Collaborated with cross-functional teams to ensure seamless integration and delivery of high-quality software.
- Participated in **daily meetings with feature leads** to ensure alignment with project standards and testing requirements.
- Key technologies used: **Python, CAPL, Vector CANoe, Bash, Jenkins, Core ALM, IBM Rational DOORS**

EDUCATION

B.Sc. in Communications and Electronics Engineering

Faculty of Engineering, Alexandria University, Egypt

2014 – 2019

CERTIFICATIONS

- **ISTQB Certified Tester Foundation Level** (Certificate ID: 221026031)
- **ISTQB Agile Tester** (Certificate ID: 230817023)
- **Modern C++ (C++11/14/17)** – Udemy (Certificate ID: UC-33d0e302-7695-4cb7-aca4-87644f7565ee)
- **Object Oriented Programming Specialization** – Coursera (Certificate ID: NGE9PDT4GKCB)

PERSONAL PROFILE

- **Date of Birth:** September 5, 1995
- **Languages:** English (Fluent), German (Basic), Arabic (Native)
- **Interests:** Open-source contributions, IoT projects, and continuous learning in software development.
- **Availability:** Open to relocation and travel as required by project needs.

REFERENCES

References are available upon request.